

DETAILED DESCRIPTION OF THE INVENTION

Multi-User Structural Matching and Compatibility Framework

In various embodiments, the encoded penile anatomical geometry profile generated by the system may be utilized within multi-user structural matching frameworks configured to evaluate geometric alignment metrics between anonymized profiles. Such matching frameworks may operate on standardized parametric descriptors rather than explicit anatomical imagery, thereby enabling quantitative comparison while preserving privacy.

A compatibility engine may compute structural alignment scores based on comparisons of longitudinal length ranges, cross-sectional radius distributions, taper gradients, curvature vectors, volumetric estimations, proportionality indices, and symmetry metrics. Similarity functions may apply weighted parameter vectors, distance metrics, multi-dimensional overlap calculations, or machine learning-based matching models trained on population-level structural datasets.

In certain embodiments, the system may generate multiple compatibility outputs reflecting different matching objectives. For example, one output may prioritize overall proportionality matching, another may prioritize curvature alignment, and another may prioritize cross-sectional distribution overlap within defined longitudinal segments. Matching outputs may be expressed as scalar scores, categorical match levels, or confidence-weighted compatibility bands.

The system may further support privacy-preserving match computation by transforming geometry profiles into condensed structural signatures comprising non-reversible feature vectors or hashed parameter sets. Such signatures may be exchanged between users or processed within a secure computation environment to generate compatibility outputs without exposing raw geometry parameters.

In some implementations, compatibility frameworks may integrate geometry-derived matching results with additional biometric or contextual measurements obtained from separate nodes within a distributed system. Composite compatibility metrics may incorporate structural alignment in conjunction with physiological synchronization indicators, temporal pattern correlation, or user-defined preference constraints.

The multi-user compatibility architectures described herein are not limited to any specific scoring method, similarity metric, signature encoding approach, or deployment environment. Any system capable of utilizing standardized penile anatomical geometry profiles to generate privacy-preserving structural matching outputs falls within the scope of the present disclosure.